

A Governed Public-Data Benchmark of Quantum Machine Learning for Human-Rated Cislunar Trajectory-Correction Planning

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Abstract. Quantum machine learning (QML) is frequently proposed for expensive physical simulation and trajectory-design problems, but a feature map or a statevector result alone does not demonstrate practical value. This paper reports a closed, governed public-data benchmark of quantum-kernel methods for human-rated cislunar trajectory-correction planning. The study couples a source-bound simulator, grouped development splits, fold-local preprocessing, physics-derived and classical controls, preregistered retention rules, and explicit evidence boundaries. The simulator credibility chain accepted 67 eligible public-data checks with zero failures. In the preregistered Gate 5 benchmark, the Q01 fidelity-style quantum kernel attained mean normalized root-mean-square error (NRMSE) 0.646614, compared with 0.008739 for the C06 physics-residual control; no regime qualified for advance. A projected quantum-kernel successor (Q01b) attained 0.661207 versus 0.006833 for C06, and a feasibility-only quantum kernel had recall 0.108860 at the frozen threshold, so neither exploratory track advanced. Sixteen subsequent, prospectively bounded development-only candidates also completed as valid negatives. The strongest apparent residual gain, ORFRK-08-R2, did not clear its matched classical-control requirement. The contribution is therefore a reproducible negative benchmark and a claim discipline: apparent QML gains in high-consequence trajectory applications must survive physics-based controls, matched dequantizations, grouped leakage protection, and frozen eligibility rules. The study does not claim quantum advantage, propellant savings, mission-loop validity, flight readiness, hardware performance, or a new QML invention.

Keywords: Quantum machine learning · quantum kernels · cislunar trajectory design · reproducible benchmarking · negative results

1 Introduction

Trajectory correction for a human-rated cislunar mission is a coupled decision problem. A correction-burn placement changes the trajectory while interacting with navigation uncertainty, maneuver execution, crew timelines, entry-interface constraints, and propellant objectives. Public Artemis trajectory-design research therefore treats robust correction cost and mission constraints together rather

than as an unconstrained curve-fitting task [14]. A learning surrogate could be useful if it reduces the cost of screening scenarios while preserving the structure that makes a candidate decision safe and interpretable. That conditional statement is not evidence that any particular machine-learning model is useful.

QML offers feature maps and hypothesis spaces that are different from common classical models [10,2]. This motivates testing QML, but it does not resolve the application question. Encoding a classical feature vector into a quantum state introduces data-loading choices, kernel geometry, and resource costs. A circuit may be difficult to simulate while still being well approximated by a classical learner on the relevant data distribution [3,13]. Conversely, a favorable numerical result can be an artifact of an under-tuned comparator, a split that leaks mission-family information, post-outcome threshold selection, or omitted failures. These are benchmarking problems before they are quantum-computing problems.

This paper documents a complete public-data experimental program for trajectory-correction prediction and feasibility screening. The purpose was not to assert a priori that QML would win. Its purpose was to establish a scientifically usable decision process: define a bounded task, verify the simulator against eligible public endpoints, freeze the development protocol, compare QML with strong controls, record negative outcomes, and prohibit claims that the evidence cannot support. This stance is consistent with model and simulation credibility practice [5] and with recent calls for fair QML comparisons [1,9].

The central research questions were deliberately narrow:

1. Can Q01 improve correction-cost prediction over physics-derived and classical controls under the frozen development protocol?
2. If Q01 fails, does a prospectively specified projected quantum kernel improve the same task without changing the data-access or split discipline?
3. Can a feasibility-only quantum kernel meet a fixed, recall-sensitive safety-filter interpretation?
4. Which conclusions remain valid when calibration, final-test, mission-loop, and hardware evidence stay locked?

The answer to the first three questions is negative for the tested models and settings. The fourth question is the paper’s main methodological contribution. The work establishes a transparent evidence hierarchy that distinguishes simulator credibility, development-only model evidence, future-design lessons, and prohibited operational claims. Negative results are not treated as failed record keeping; they are the result when a hypothesis does not meet its predeclared standard.

The paper makes five contributions:

1. A source-bound cislunar benchmark protocol with public inputs, grouped development folds, fold-local transformations, and retained numerical feasibility outcomes.
2. A simulator credibility chain that reports its eligible scope and its source-ineligible exception rather than converting missing evidence into a pass or failure.

Table 1. Claim boundary used throughout the manuscript. This table is derived from the committed claim-boundary record and does not add a new experimental result.

Evidence category	Permitted interpretation
Gate 3	Simulator agreement for the eligible public-data checks; not flight validation. RTC3 is source-ineligible and is neither a pass nor a failure.
Gate 5	Under the frozen public-data development benchmark, tested QML candidates did not outperform strong classical controls.
P001 and D034–D049	Development-only exploratory or successor negatives. They do not revise Gate 5 or authorize a Gate 6 run.
Safety lesson	The recall-first classical analysis is a future protocol-design lesson, not a deployed safety filter.
Prohibited claims	No quantum advantage, propellant saving, mission-loop validity, flight readiness, NASA approval, hardware behavior, final-test result, or QML invention claim.

3. A preregistered QML comparison against physics-residual, random-feature, compressed-input, and classical kernel controls.
4. A closed successor ledger containing exploratory and prospectively bounded negative candidates, including matched classical dequantization controls.
5. A self-contained evidence and figure package that supports review of the negative conclusion without widening the study’s scientific claim.

2 Scientific Scope and Claim Boundary

The study is an offline, public-data, development-only benchmark. It is not a mission design authority, a flight certification exercise, or a performance assessment of a quantum processor. A simulation result may be useful for evaluating a model family, but it cannot by itself establish the performance of a mission-owned trajectory system or a hardware quantum implementation. The distinction is essential for a human-rated application, where human-system requirements are separate from a surrogate’s regression score [6].

The evidence boundary was defined before reporting. Gate 3 supports simulator credibility only for the eligible comparisons. Gate 5 supports a technical decision for a frozen development benchmark. The subsequent P001 and D034–D049 branches are development-only follow-ups; they cannot retroactively change the Gate 5 decision. Calibration and final-test rows, mission-loop scenarios, hardware and GPU quantum execution, threshold application to real operations, trained-model release, and Gate 6 remain outside the study. Table 1 summarizes the resulting language constraints.

Figure 1 records the governance state rather than a model metric. It is included because a benchmark conclusion depends on which evidence was authorized when. A later candidate may be scientifically informative, but it does not

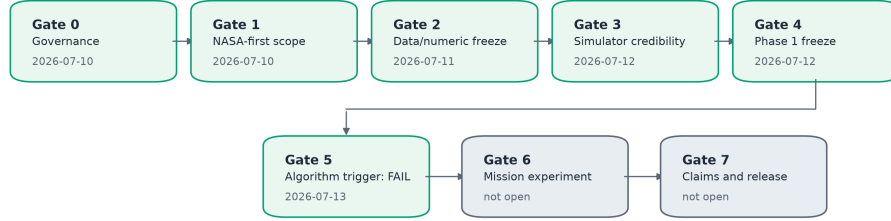
Research governance decision diagram through Gate 5

Diagram only: dates and decision states do not imply scientific success.

Fig. 1. Research governance through Gate 5 (RFIG-001). Gates 0–5 were closed by human decisions; the technical Gate 5 result is a benchmark-specific FAIL. Gate 6 remains unopened. This is decision-state evidence, not a model performance plot.

become preregistered evidence merely because it is reported after a negative primary result.

3 Related Work and Positioning

The target problem is positioned relative to public work on Artemis trajectory-correction design, rather than being presented as a replacement for mission-owned analysis. Woffinden et al. describe a correction-burn placement formulation that incorporates uncertainty, crew-related timing, and entry constraints [14]. Classical learning research has explored low-thrust guidance and cislunar transfer design, but those settings differ in dynamics, propulsion, objectives, and validation structure [12,4,15]. They are valuable precedents for surrogate modeling and for retaining infeasible cases, not interchangeable baselines for a human-rated correction benchmark.

Quantum feature-space methods express a classical input through an encoding and compare encoded states by a quantum-derived similarity [10,2]. The method can be used in a kernel learner, but its predictive value depends on an alignment between the data geometry, the feature map, the measurement, and the task. Theoretical and numerical work has made this caveat increasingly explicit: hard-to-simulate circuits do not automatically give a learning advantage, and the data distribution can make a classical approximation competitive [3,13].

The present work is particularly informed by three benchmarking lessons. First, kernel bandwidth and encoding scale are part of the effective inductive bias, not negligible implementation details [11]. Second, a quantum-kernel study should include appropriately strong classical counterparts, including random features where relevant [7,13]. Third, a comparison should report more than its best

run: model choice, split structure, seed variation, capacity, and post-outcome choices all affect what a claimed gain means [1,9,8].

This paper does not make an exhaustive literature-review claim. The repository preserves a bounded extraction set of 23 accepted records used to establish the initial protocol and a separate archival discovery refresh. The refresh did not change the experiment, thresholds, figures, or outcomes. Literature is therefore used here to position the benchmark and justify controls, not to construct a retrospective theory of success.

4 Materials and Methods

4.1 Scenario construction and simulator credibility

The public pipeline generated correction scenarios over the frozen F0, F1, and F2 fidelity levels. The implementation retained mass depletion, event handling, crew-related constraints, uncertainty variables, and numerical feasibility checks. The intent was to predict benchmark-defined cost or feasibility outcomes, not to control a spacecraft. The source and scenario construction were frozen before the model campaigns; later reports could summarize the outputs but could not replace source evidence to make a comparison pass.

Simulator credibility was assessed in Gate 3 by comparing eligible Python endpoints with GMAT endpoints under the accepted force-model configurations. The diagnostic chain isolated an Earth-J2 coefficient-file formatting issue and confirmed the repaired format through standard degree-two, order-zero comparisons. In the final accepted evidence, all 67 evaluable checks passed; the repaired endpoint set also passed the frozen position and velocity limits. This verifies an implementation boundary, not a full mission environment.

One record, RTC3, is explicitly excluded from the claim. Its public qualified source predates the event under the frozen temporal-evidence rule. It is therefore classified as `not_eligible`, rather than as a successful or failed endpoint. This distinction prevents an important category error: source eligibility is not a numerical outcome. The exclusion remains visible in the evidence package rather than being silently omitted.

4.2 Frozen development data and leakage control

Let $\mathcal{D} = \{(x_i, y_i, z_i, g_i)\}_{i=1}^N$ denote the frozen development dataset. Here x_i is the scenario feature vector, y_i is the robust correction-cost target, z_i is the independently propagated feasibility label when that endpoint is used, and g_i is a mission-design group. Groups, rather than individual records, are assigned to development folds. Thus records sharing a mission-design group do not appear on opposite sides of a training-validation partition.

All transformations are learned within the training partition of each fold. This includes imputation, categorical encoding, scaling, target scaling, and principal-component analysis (PCA) where configured. The requirement is important

because globally fitted scaling or PCA can transmit held-out structure into a training representation even when the row identifiers are not directly used. The procedure makes the reported errors more conservative, but it makes a model comparison interpretable.

The development campaign uses nested hash-selected sample rungs of 128, 256, 512, and 1,024 rows. Selected configurations are evaluated over 20 frozen seed indices. These are not attempts to estimate an operational distribution; they are a prespecified stress test for whether a candidate can clear its development advancement rule across retained group structure and controlled randomness. Calibration and final-test data were never read by these campaigns.

4.3 Task definitions and estimands

The primary endpoint is robust correction-cost prediction. For a fitted predictor f , a fold-level error takes the form

$$\text{NRMSE}(f) = \frac{1}{s_y} \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2}, \quad (1)$$

where s_y is the frozen target scale for the corresponding evaluation configuration. The primary report uses pooled out-of-fold NRMSE, while unweighted fold summaries and seed-level diagnostics are retained to expose heterogeneity. Robust correction cost and feasibility-constrained regret are interpretation metrics rather than substitutes for the frozen primary endpoint.

The classification track estimates a feasibility score $p_f(x)$. It is evaluated with Brier score, AUROC, recall, precision, false-negative rate, and calibration diagnostics. For a fixed threshold $t = 0.5$, recall is

$$\text{Recall}_{0.5} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (2)$$

This endpoint has a different scientific role from regression. A feasibility model with adequate discrimination but poor recall can be unsuitable as a safety-filter candidate even if its AUROC appears reasonable. The study keeps these interpretations separate.

The Q01 family is described generically by an encoded-state kernel

$$K_q(x, x') = \left| \langle 0 | U_\phi(x)^\dagger U_\phi(x') | 0 \rangle \right|^2, \quad (3)$$

where U_ϕ denotes the frozen feature-map circuit. Equation (3) states the family-level fidelity construction; the exact feature, PCA, bandwidth, and regularization settings are held in the committed configuration records. A kernel learner maps the resulting Gram matrix to a prediction. This representation is not itself a claim of a hardware implementation or a quantum speedup.

4.4 Models and matched controls

The preregistered Q01 candidate is a fidelity-style quantum-kernel regressor for the correction-cost endpoint. Its principal comparator is C06, a physics-residual control that combines a low-fidelity physical baseline with a classical residual learner. C06 is a more demanding comparator than a generic regressor because it encodes useful problem structure available to both the scenario and the benchmark. The registered comparison set also contains A01 random Fourier features, compressed classical views, and other frozen classical-family configurations.

After the accepted Gate 5 result, P001 defined two prospective exploratory tracks without reopening the primary campaign. Q01b uses reduced-density matrix projections and a median-distance projected kernel. FQK applies a quantum kernel only to the independently propagated feasibility label. Both tracks retained groupwise splits, fold-local preprocessing, nested rungs, matched controls, and 20 seed indices. They were allowed to produce a new development-only signal, but not to revise the decision already made for Q01.

The D034–D049 successor branch tightened this discipline further. Each protocol froze its candidate, target, comparator, threshold, stopping rule, and source path before fitting. Several candidates produced a lower NRMSE than C06. That alone was insufficient. A candidate had to survive its matched classical alternative and branch-specific rule, so a gain attributable to a classical residual construction could not be labeled quantum-specific.

4.5 Statistical decision process

The preregistered Gate 5 trigger required complete eligible folds and seeds, comparison with the strongest classical finalist and matched dequantization controls, paired uncertainty checks, and a qualifying regime defined before outcomes were observed. The reporting package retains paired bootstrap intervals, fold-level sign diagnostics, and a sign-permutation test with multiplicity control. It does not add a post-outcome seed, model, split, or threshold to repair a weak result.

The distinction between task completion and eligibility is also retained. D006 completed 871 of 871 authorized tasks with zero task failures. Q02 and Q03 were not carried to later rungs because they did not meet the frozen retention rule. Their status is therefore `not_reached_under_frozen_eligibility`. It does not mean a failed experiment, a missing computation, or a favorable result withheld from the report.

4.6 Computational-resource and reproducibility boundary

The reference compute environment was a Windows laptop with a 13th Generation Intel Core i9-13900HX CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4060 Laptop GPU. The experiment applied a bounded-RAM and serialized-statevector policy; the GPU and quantum hardware were not used for the reported QML evidence. The hardware description is included so that readers understand the throughput and resource constraints under which the public

benchmark was executed. It is not a prescription that future studies use identical hardware.

The source release contains configuration records, seeds, data cards, figure registry entries, machine-readable tables, and reporting scripts. Every figure used in this paper is copied with its corresponding registry entry into the paper package. These measures make the evidence auditable while preserving the lock on calibration and final-test data.

5 Results

5.1 Simulator credibility result

The accepted Gate 3 record contains 67 evaluable checks and zero failures. The result includes the Earth-only diagnostic chain that localized the original J2 discrepancy to a coefficient-file delimiter problem rather than to the integrator, time system, or initial state. Once the coefficient format was repaired, the GMAT-Python comparison passed the unchanged position and velocity thresholds across the eligible windows. The appropriate conclusion is that the public simulator implements the tested scope consistently enough to support the development benchmark. It is not a flight-readiness conclusion, and RTC3 remains source-ineligible.

5.2 Preregistered Gate 5 benchmark

The Q01 quantum kernel did not satisfy the development trigger. Its mean NRMSE was 0.646614, whereas C06 achieved 0.008739. The difference is sufficiently large that it cannot be interpreted as a close race resolved by one uncertainty statistic. No residual regime satisfied the frozen advance rule. Gate 5 is therefore a valid technical FAIL for the Q01 model, task, preprocessing, controls, and development population specified by the protocol.

Figure 2 shows the pooled development evidence over the authorized rungs and matched PCA dimensions. Configurations that did not advance were stopped at their retention gate. Figure 3 reports the selected-model distribution over the 20 frozen seed indices, and Fig. 4 places the trigger next to its strongest classical and tuned dequantization controls. The three figures should be read together: a model is not evaluated solely by a single aggregate mean, but by its retained trajectory through the precommitted decision rules.

Q02 and Q03 require a precise reading. They completed their authorized tasks, but they did not satisfy the frozen eligibility condition for later rungs and seed reruns. They are recorded as `not_reached_under_frozen_eligibility`. Treating this state as a failed experiment would overstate the evidence; treating it as an unreported potential success would ignore the protocol. The paper reports the actual decision state.

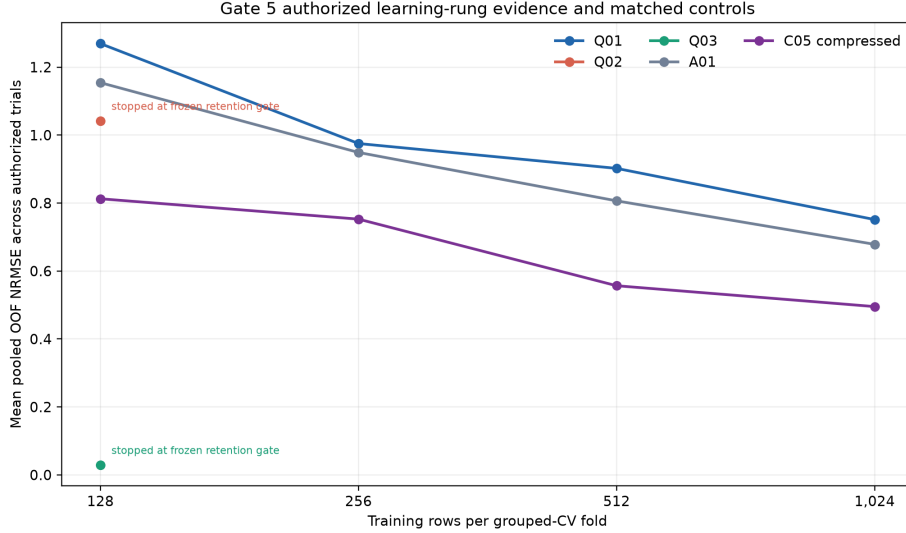


Fig. 2. Gate 5 development learning curves and matched controls (RFIG-021). The evidence covers every authorized rung and matched PCA dimension. This is development model-selection evidence only.

5.3 Post-Gate 5 exploratory tracks

Q01b tested whether changing the quantum readout while keeping the same development discipline could materially alter the result. It did not. Q01b attained mean NRMSE 0.661207, compared with 0.006833 for C06, a relative gap of 95.77 times. Figure 5 presents its learning curves against the source-matched controls, while Fig. 6 reports the projected-kernel geometry and dequantization diagnostics. These are useful negative data: for this projected feature map, the representation did not capture the residual structure better than the classical controls. They do not establish that projected quantum kernels cannot work on any other physical problem.

The FQK classification track separates discrimination from a safety-filter interpretation. It achieved AUROC 0.7436 and Brier score 0.1561, but recall at the fixed threshold was only 0.1089. In a setting where missed unsafe cases are material, the latter outcome prevents promotion to a safety-filter signal. The endpoint plot in Fig. 7 makes this distinction visible. It is not valid to call FQK useful for safety merely because one scalar metric is above chance or because a quantum feature map was evaluated.

The recall-first CSAFE-RF analysis provides a separate lesson for a future protocol. A calibrated logistic model reached mean recall 0.804253 with false-negative rate 0.195747 and Brier score 0.142231. That result identifies an objective-design tension: a model selected only for calibration can miss an unacceptable number of unsafe cases, whereas a future safety protocol should prospec-

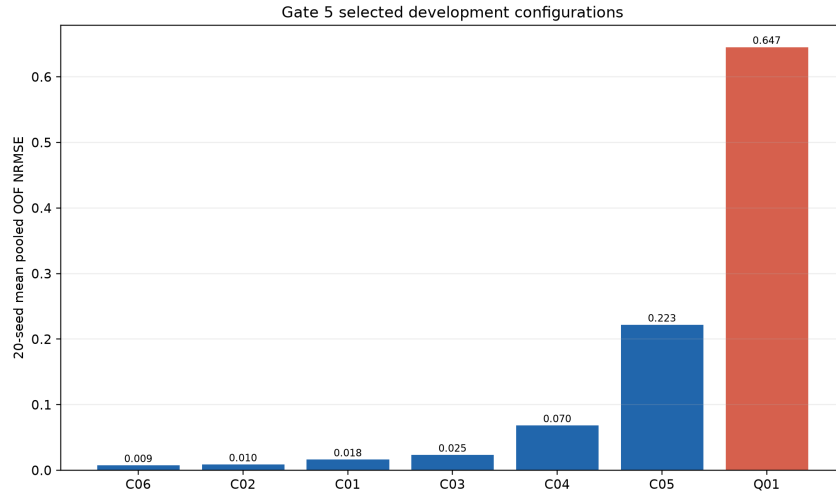


Fig. 3. Twenty-seed summary of selected Gate 5 models (RFIG-022). Seed variation is reported as a stability diagnostic, not as final-test evidence.

tively define recall and calibration constraints together. CSAFE-RF was not used to re-rank QML candidates, change a threshold, or authorize a mission filter.

5.4 Closed successor branch D034–D049

The closed successor branch contains 16 prospectively bounded candidates. All read the development scope only, used 39,000 development rows per campaign, and completed with the recorded status `complete_valid_negative`. The complete results are listed in Table 3. The table is intentionally included in full rather than selecting only candidates with lower raw error than C06.

Several variants have a lower mean NRMSE than C06. This is a reason to inspect the controls, not a reason to announce a quantum-specific result. The best raw C06 comparison was D046, ORFRK-08-R2: NRMSE 0.0064363 versus 0.0068328 for C06, an apparent 5.80% reduction. Its matched second-stage RBF control, TWO-RBF, attained 0.0067501. The candidate’s improvement over that matched classical alternative was therefore about 4.65%, which is below the frozen 5% branch-specific rule. The result does not meet the evidence standard for a quantum-specific candidate.

No regime cell satisfied every preregistered condition

Preregistered Gate 5 trigger: FAIL. Calibration reads = 0; final-test reads = 0.

Fig. 4. Preregistered algorithm-trigger regime controls (RFIG-023). A passing development trigger would have supported consideration of one further algorithm study, not quantum advantage or operational deployment.

Table 3: Closed D034–D049 successor branch. Every row completed as a valid development-only negative. The C06 NRMSE was 0.006833 for each row. A negative paired difference favors the candidate over C06, but each branch also required its own frozen comparison and stopping rules.

Decision	Candidate	Candidate NRMSE	Difference C06	vs 95% paired interval
D034	PRQK-08-N	0.029326	0.022493	[0.022486, 0.022500]
D035	CFQSR-08-N	0.013463	0.006630	[0.006627, 0.006632]
D036	TAPQK-08	0.010382	0.003549	[0.003547, 0.003551]
D037	TSQR-08-L025	0.006848	0.000015	[0.000015, 0.000015]
D038	GFRK-08-L010	0.006647	-0.000186	[-0.000187, -0.000184]
D039	EC-GFRK-08-L010	0.006466	-0.000366	[-0.000368, -0.000365]
D040	CE-GFRK-08-L010	0.006504	-0.000329	[-0.000330, -0.000328]
D041	HEFRK-08-E050	0.006592	-0.000241	[-0.000242, -0.000240]
D042	AGEFRK-08-ADAPT	0.006524	-0.000309	[-0.000310, -0.000308]
D043	SFRK-08-CV	0.006445	-0.000387	[-0.000389, -0.000386]
D044	NIFRK-08-NL	0.006751	-0.000081	[-0.000084, -0.000079]
D045	MSFRK-ALL	0.006731	-0.000102	[-0.000103, -0.000100]
D046	ORFRK-08-R2	0.006436	-0.000397	[-0.000399, -0.000394]
D047	OMFRK-ALL	0.006952	0.000119	[0.000113, 0.000125]
D048	NSORFRK-08	0.006590	-0.000243	[-0.000244, -0.000242]
D049	SSORFRK-08	0.006566	-0.000266	[-0.000268, -0.000265]

5.5 Evidence accounting and closure

Evidence accounting confirms that the reported conclusion follows the authorized boundary. D006 completed 871 of 871 tasks, with zero task failures and zero reads of calibration or final-test data. The D011-R1 exploratory campaign read 39,000 development rows and zero locked rows. No hardware quantum job, mission-loop run, Gate 6 run, or final-test evaluation occurred.

Table 2. Primary and exploratory QML endpoints from the frozen evidence package. All values are development-only. The feasibility endpoint uses the frozen 0.5 decision threshold.

Block	Candidate and endpoint	Value and comparator	Interpretation
Gate 5	Q01 quantum kernel; mean NRMSE	0.646614 vs 0.008739	C06: Technical FAIL; no qualifying regime
P001	Q01b projected quantum kernel; mean NRMSE	0.661207 vs 0.006833	C06: Exploratory negative; 95.77x relative gap
P001	FQK feasibility kernel; recall at 0.5	0.108860; controls stronger	classical Exploratory negative; not a safety filter
Future lesson	Calibrated logistic; recall / Brier	0.804253 / 0.142231	/ Protocol-design signal only

Figure 10 summarizes the closure recommendation. It reports that no candidate is eligible for Gate 6; it does not represent an unexecuted Gate 6 experiment. The experimental program was subsequently closed under D055-C to permit publication preparation without turning future protocol ideas into unapproved computation.

6 Discussion

The principal finding is a controlled negative benchmark. The Q01 fidelity-style quantum kernel was far behind the physics-residual control, and the Q01b projected feature map did not close the gap. The feasibility-only kernel did not meet the recall-sensitive safety interpretation. The successor branch demonstrates an additional failure mode that is important beyond this specific task: a candidate can improve a weak or incomplete comparator yet fail against a matched classical construction. That distinction is exactly why dequantization controls and branch-specific thresholds belong in the protocol.

The result is scientifically useful because the negative conclusion is specific. It rules out a family of tested feature maps and successor constructions under a particular public-data scenario generator, grouped development procedure, resource envelope, and comparator set. It does not rule out all QML, all cislunar machine learning, or all future quantum hardware. Such broad claims would require evidence absent here, including separate tasks, new preregistration, scalability studies, noise analysis, hardware measurements, and appropriate classical resource comparisons.

The physics-residual control is a key methodological result. A generic classical baseline is not sufficient when the prediction target contains structure that a low-fidelity model can represent directly. C06’s performance shows that, for this benchmark, the available physical residual structure is more informative than the tested quantum fidelity features. That observation does not establish a theorem

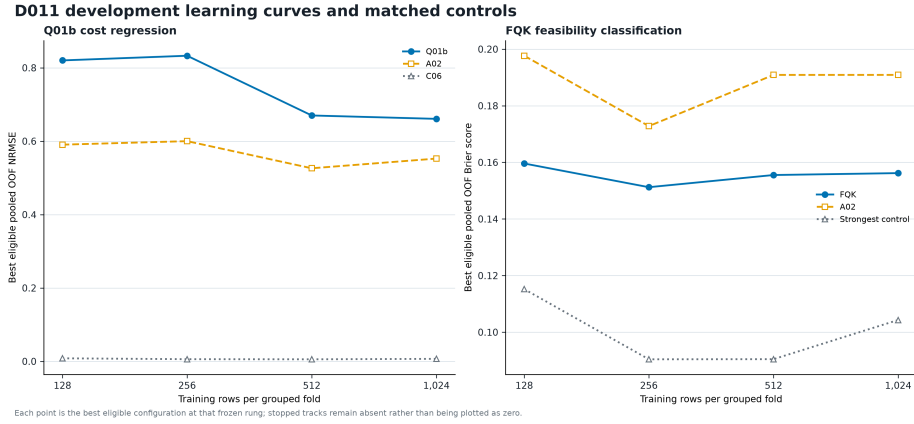


Fig. 5. D011/P001 projected-kernel learning curves and matched controls (RFIG-026). The figure is development-only exploratory evidence and does not revise the Gate 5 decision.

about quantum circuits; it establishes a task-specific inductive-bias comparison that future proposals must beat fairly.

The retained Q02/Q03 status also matters. Many reports collapse all non-advancing paths into a single “failure” category. Here, a task can run correctly but be ineligible for later computation under a rule frozen before results. Keeping `not_reached_under_frozen_eligibility` separate from a task failure makes the evidence ledger more truthful and prevents a later analyst from inventing a selection path that never existed.

The paper’s effect is therefore not a claim of operational benefit. It is a reusable scientific pattern for high-consequence QML evaluation: include an eligible simulator-validation scope, freeze data transformations and groups, use a physics-aware control, compare against matched classical alternatives, retain null outcomes, separate future protocol lessons from results, and use claim boundaries that are mechanically checkable from source artifacts. This pattern is valuable precisely because it makes an exciting but unsupported result harder to publish.

6.1 Implications for QML benchmarking

The QML literature has repeatedly shown that benchmark conclusions depend on which classical methods are made available to the comparison [1,9]. The D046 result gives a concrete trajectory application example. Comparing only with C06 would make the candidate’s raw 5.80% difference look promising. The matched TWO-RBF alternative explains enough of that difference that the candidate fails the frozen 5% rule. The interpretation changes because the question changes from “is the candidate better than one baseline?” to “is its incremental mechanism still supported when a classical model is given the same residual construction?”

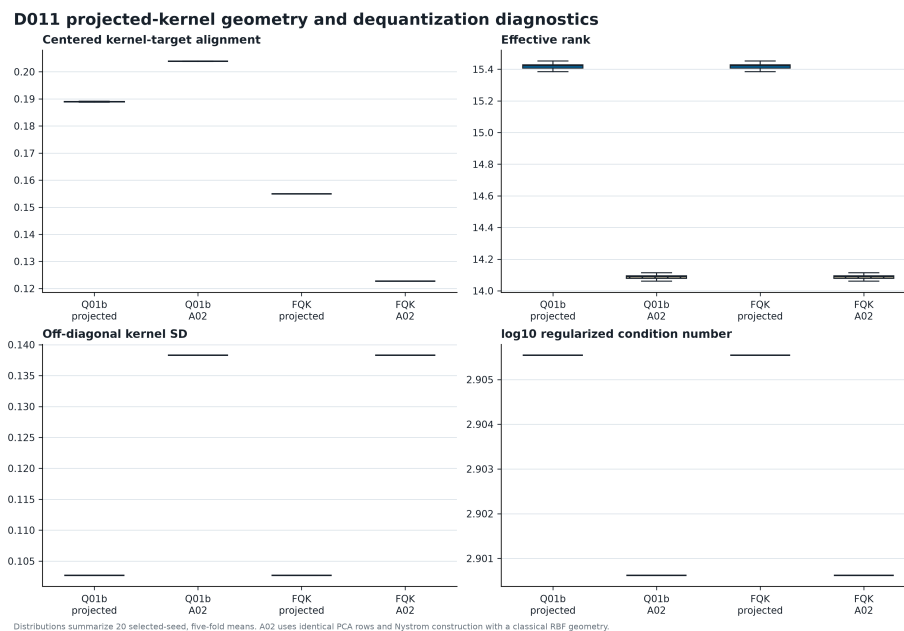


Fig. 6. Projected-kernel geometry and dequantization diagnostics (RFIG-027). Selected projected kernels are compared with A02 classical RBF geometry on the same fold-local PCA rows and Nystrom construction.

This contrast does not show that the matched control is the universally best classical model. It shows why a QML result needs multiple controls chosen to test the claimed mechanism. A fair comparison is not one that intentionally makes every model identical; it is one that removes alternative explanations for the purported QML contribution. In this benchmark, the alternative explanation was a classical residual representation, and it remained competitive.

6.2 Safety objective as a distinct scientific question

The FQK and CSAFE-RF records demonstrate that predictive metrics must be tied to the intended decision. AUROC and Brier score summarize useful aspects of a classifier, but neither alone ensures that the model detects enough unsafe cases at a fixed threshold. The FQK result therefore remains negative under the study’s safety-filter interpretation even though its discrimination score is above random. This guards against a common reporting error: promoting a model by the metric that is most favorable after the outcome is known.

The higher-recall classical result is not a rescue or a replacement result. It identifies a future hypothesis: a safety screen should define a recall floor, false-negative constraint, calibration requirement, and action policy before the model is fitted. That hypothesis belongs to a separately authorized experiment, because

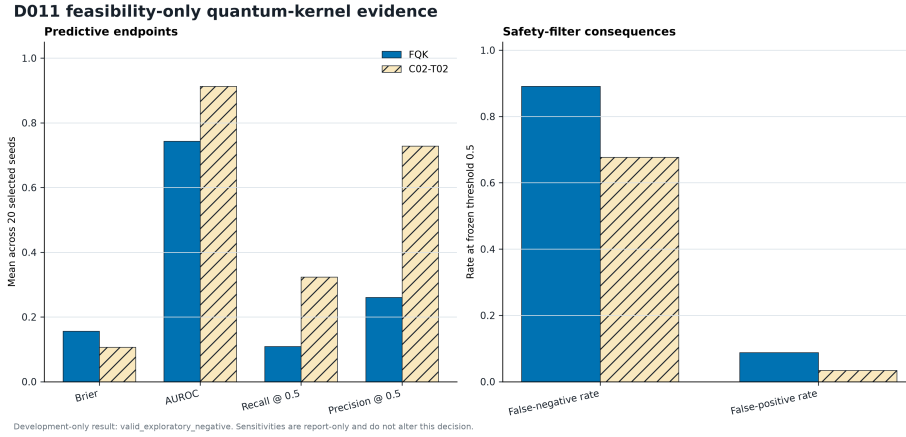


Fig. 7. FQK predictive and safety-filter endpoints (RFIG-028). Brier, discrimination, recall, precision, and consequence metrics are compared with the strongest complete control. The frozen recall interpretation did not pass.

using it to alter the Gate 5 rule after inspection would invalidate the original test.

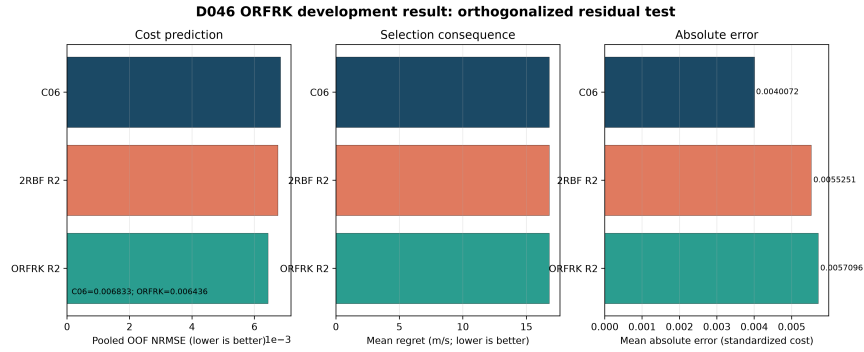
7 Limitations and Threats to Validity

First, the study is public-data and development-only. Calibration and final-test data were not read. The paper cannot estimate final-test error, operational calibration, mission-loop behavior, propellant savings, or flight readiness. The use of Artemis-related public work establishes relevance of the problem framing, not access to mission-owned truth data or authority to draw mission conclusions.

Second, simulator agreement is bounded. The Gate 3 results support the eligible public endpoint comparisons, but they do not validate every force model, contingency, navigation product, or human-rating consideration. RTC3 is source-ineligible, and some generated scenario decisions do not have an independently feasible numerical reference. Those limitations remain in the dataset rather than being removed after model results are observed.

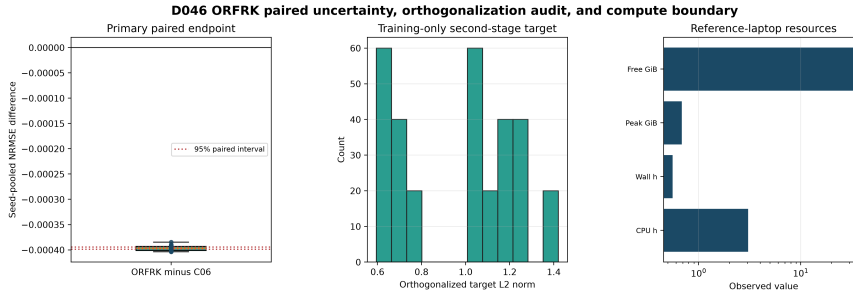
Third, the quantum experiments are small-qubit classical statevector simulations within a laptop-bounded resource envelope. They do not report shot noise, gate noise, device connectivity, state-preparation latency, transpilation cost, hardware calibration, or end-to-end quantum runtime. They cannot demonstrate quantum speedup, hardware efficiency, or a practical quantum advantage. An alleged advantage would require all of those questions to be addressed against a fair classical resource baseline.

Fourth, the methods are limited by their frozen design. The chosen preprocessing, feature maps, sample rungs, controls, and thresholds make the comparison interpretable but restrict the scope of inference. A different method can be



Five grouped outer folds; four inner whole-group folds; 20 seeds; 39,000 development rows. The first-stage RBF-0.25 correction is shared by candidate and control.

Fig. 8. D046 ORFRK result comparison (RFIG-078). The $q=8$ candidate and a matched second-stage RBF control are compared after a shared RBF-0.25 correction. Lower raw error than C06 is not sufficient for a QML claim.



Development-only evidence; outer validation labels were used only for scoring. Calibration/final-test, hardware/GPU, mission, and Gate 6 counters were zero.

Fig. 9. D046 paired audit and resource boundary (RFIG-079). The figure reports the paired endpoint, training-only orthogonalization checks, integrity evidence, and reference-laptop resources. It is not hardware or GPU evidence.

scientifically tested only through a new protocol, not by modifying the completed campaign until a favorable result appears.

Finally, the literature synthesis is bounded, not systematic. The inclusion set provides methodological context and control requirements but does not prove that no related method exists. This is why the paper makes a benchmark claim rather than a novelty or universal-performance claim.

8 Future Research Boundary

The negative results motivate questions about task-specific symmetry, representation, and residual structure, but the completed evidence does not answer them. A future geometric-QML direction was discussed as a hypothesis, not as

D025-C Gate 5 closure: no QML candidate for Gate 6

All Gate 5 and post-Gate-5 evidence is development-only or reporting-only; Gate 6 remains locked.

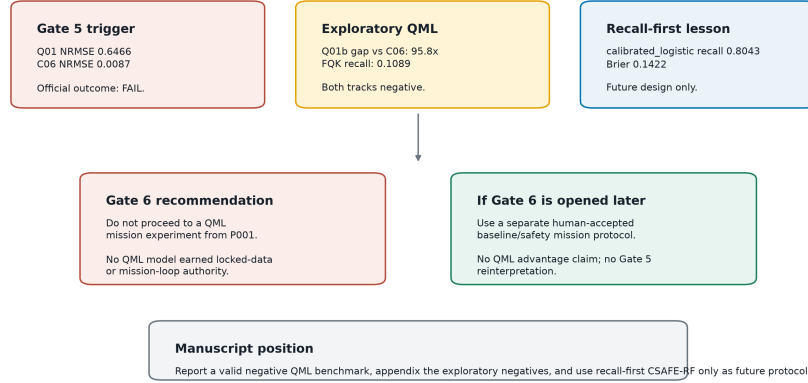


Fig. 10. Gate 5 closure and Gate 6 recommendation (RFIG-044). No QML candidate from the closed evidence branches is eligible for Gate 6. This is a closure recommendation, not a Gate 6 result.

a result: determine whether any transformation preserves the actual cislunar inputs and labels in the presence of fixed Earth–Moon–Sun ephemerides, thrust constraints, and reference-frame choices. Equivariance alone would not imply a QML advantage over classical equivariant models.

A future protocol would require a mathematical contract, a symmetry audit, baseline reproduction, a representation- and parameter-matched classical comparison, ablations of symmetry and initialization, resource accounting, and explicit rejection criteria. The D054-A/P018 binding freeze records how a future audit could specify its representation, DLA basis, generators, development-only strata, tolerance, output paths, and figures. D055-C declined to execute that audit. Accordingly, this manuscript contains no DOP853/SVD audit result, no new symmetry dataset, no variational circuit training, and no geometric-QML performance claim.

This boundary is not a weakness in reporting. It prevents a promising idea from being laundered into evidence through narrative. The appropriate next step, if future authors choose to reopen research, is a separately approved protocol that can reject the hypothesis as readily as it can support it.

9 Data, Code, and Figure Availability

The source repository, governance records, machine-readable reporting outputs, figure registry, and the published source release are available at the project repository. The published release record is release v0.3.0. This paper package copies the exact figures used in the manuscript and the three core result tables

plus figure registry. The package does not include calibration or final-test rows, mission-owned evidence, trained weights, or unexecuted future-protocol outputs.

10 Conclusion

Under the frozen public-data development benchmark, the tested QML candidates did not outperform strong classical controls, and no candidate from the Gate 5 or D034–D049 branches is eligible for a Gate 6 mission experiment. The primary Q01 quantum kernel, the projected Q01b kernel, and the feasibility-only FQK track all produced negative outcomes under their stated criteria. The strongest successor improvement over C06 did not persist above its matched classical-control threshold.

The study’s contribution is a reproducible negative result with an explicit governance and claim structure. It reports what was tested, what did not advance, what is merely a future lesson, and what is outside the evidence. It does not establish quantum advantage, fuel savings, mission utility, hardware performance, flight readiness, NASA approval, or an invented QML model. That restraint is necessary for the result to be useful.

Appendix: Reproducibility and Evidence Ledger

The paper package is structured so that a reviewer can trace every main-text number to a copied source table or figure. The Gate 5 CSV and the successor branch-summary CSV in `results_tables/` provide the numerical values reported in Tables 2 and 3. The copied figure registry records the title, evidence status, source data, generator, digest, and claim boundary for every figure included here. The figures retain their registry identifiers in their captions.

The release package provides scripts, fixed configuration records, data cards, and decision logs. Reproduction of the published development figures should begin from the tagged repository state and should preserve the lock on calibration and final-test data. Re-running a report-generation script is different from authorizing new fitting or a new experimental campaign.

Interpretation Checklist for Reviewers

The following checklist summarizes the intended reading of the paper:

1. Treat Gate 3 as an eligible-scope simulator consistency result, not a mission validation result.
2. Treat Gate 5 as a preregistered development benchmark FAIL for Q01, not a universal statement that QML cannot work.
3. Treat Q01b, FQK, and D034–D049 as development-only negative evidence that cannot revise Gate 5 or open Gate 6.
4. Treat the D046 result as evidence that a raw improvement over C06 must be tested against a matched classical control.

5. Treat the recall-first analysis as future protocol-design evidence, not as a deployed safety filter.
6. Do not infer quantum advantage, hardware performance, propellant saving, or flight readiness from any figure or table in this package.

References

1. Bowles, J., Ahmed, S., Schuld, M.: Better than classical? the subtle art of benchmarking quantum machine learning models (2024), <https://arxiv.org/abs/2403.07059>
2. Havlicek, V., et al.: Supervised learning with quantum-enhanced feature spaces. *Nature* **567**, 209–212 (2019). <https://doi.org/10.1038/s41586-019-0980-2>
3. Huang, H.Y., et al.: Power of data in quantum machine learning. *Nature Communications* **12**, 2631 (2021). <https://doi.org/10.1038/s41467-021-22539-9>
4. Izzo, D., et al.: Real-time guidance for low-thrust transfers using deep artificial neural networks. *Journal of Guidance, Control, and Dynamics* (2021). <https://doi.org/10.2514/1.G005254>
5. National Aeronautics and Space Administration: NASA-STD-7009B: Standard for models and simulations. Tech. rep., NASA (2024)
6. National Aeronautics and Space Administration: NASA-STD-3001, volume 2, revision e: Human factors, habitability, and environmental health. Tech. rep., NASA (2025)
7. Otten, M., et al.: Quantum machine learning using gaussian processes with performant quantum kernels (2020), <https://arxiv.org/abs/2004.11280>
8. Pernot, P., Huang, J., Savin, A.: Impact of non-normal error distributions on the benchmarking and ranking of quantum machine learning models. *Machine Learning: Science and Technology* **1**(3), 035001 (2020). <https://doi.org/10.1088/2632-2153/aba184>
9. Schnabel, P., Roth, I.: Quantum kernel methods under scrutiny: A benchmarking study (2024), <https://arxiv.org/abs/2409.04406>
10. Schuld, M., Killoran, N.: Quantum machine learning in feature hilbert spaces. *Physical Review Letters* **122**(4), 040504 (2019). <https://doi.org/10.1103/PhysRevLett.122.040504>
11. Shaydulin, R., Wild, S.M.: Importance of kernel bandwidth in quantum machine learning. *Physical Review A* **106**, 042407 (2022). <https://doi.org/10.1103/PhysRevA.106.042407>
12. Sullivan, C.J., et al.: Multi-objective reinforcement learning for low-thrust transfer design between libration point orbits (2021), https://ntrs.nasa.gov/api/citations/20210018983/downloads/Multi_Objective_Reinforcement_Learning_for_Low_Thrust_Transfer_Design_between_Libration_Point_Orbits.pdf
13. Sweke, R., et al.: Potential and limitations of random fourier features for dequantizing quantum machine learning. *Quantum* (2025). <https://doi.org/10.22331/q-2025-02-20-1640>
14. Woffinden, D., Eckman, A., Robinson, S.: Optimized trajectory correction burn placement for the NASA artemis II mission. Tech. rep., NASA (2023), <https://ntrs.nasa.gov/api/citations/20230000223/downloads/Artemis20ptTrajDesignFinal.pdf>
15. Xie, Y., Dempster, A.G.: Feasible low-thrust trajectory identification via a deep neural network classifier (2022), <https://arxiv.org/abs/2202.04962>